EC 432

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Homework 2

Inflation forecasts, Monte Carlo Experiments, Lag Polynomials, Stationarity, Geometric Brwonian Motion

Due to 14/03/2025

i) Read lecture notes. Derive γ_k for the random walk with a drift, AR (1) and MA(1).

ii) Collect Inflation monthly data from TCMB. http://www.tcmb.gov.tr

iii) Check for stationarity using unit root tests such as ADF.

iv.) De-seasonalize it. You can use any method you like. Also record the seasonal factor.

v.) Choose a model (any ARIMA model) and determine the optimal number of lags using Box-Jenkins Approach. Yo can use Akaike Information Criterion. (AIC)

vi.) Forecast March and April inflations

vii) Forecast March and April inflations using $AR(1)$, $MA(1)$, $ARMA(1,1)$ and $AR(2)$ model. Do not forget to feed back the seasonal adjustment that you have calculated in iv).

viii) Simulate AR(1), MA(1), Random Walk, Random Walk with a drift ARIMA(1,1,1)Choose parameters such that to show different types of dynamics. Plot in the same graphHint use arima.sim() function in R. Remember to set seed first.

ix) Find out if the following processes are **covariance stationary**

$Y_t = (0.8 + 1.8L + 0.6L^2)\epsilon_t$ where ϵ_t are i.i.d with $N(\mu, \sigma^2)$

$$(1-0.2L-0.1L^2)Y_t=\epsilon_t \text{ where } E(\epsilon_t,\epsilon_\tau)=\begin{cases}1, & \text{for } t=\tau\\ 0, & \text{otherwise}\end{cases}$$

x) Find the **lag polynomial** that converts the following processes into an MA process
i)$Y_t = 0.1Y_{t-1} + \epsilon_t$
ii)$Y_t = 1 + 0.3Y_{t-1} + \epsilon_t$

xi) Design a **Monte Carlo experiment** on the $AR(1)$ process with $\alpha = 0.58$ and $\beta = 0.67$ and $i = 300$ iterations to show small sample bias of OLS. Report your statistics: compare mean parameter values obtained from the experiment vs. true parameter values in a graph where the x-axis shows the number of iterations and the y axis shows i) the mean parameter values for each iteration and ii) their true values. In another graph with the same x-axis show i) the mean square error of the residuals, where
$MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i\text{-}\mathrm{Y}_i)^2$and ii) mean square error of the parameters.

xii) . Pick a stock exchange. Collect its index using one of the API options available. Calculate σ and α for a **Geometric Brownian motion with a Drift to simulate monthly**.